

# KUNDANA RAO

**AI/ML ENGINEER | MACHINE LEARNING | GENERATIVE AI | MLOPS**

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<https://kundana2508.github.io/Kundana-portfolio/>

## PROFESSIONAL SUMMARY

AI/ML Engineer with 3+ years of experience building and deploying machine learning and Generative AI solutions across the end-to-end ML lifecycle. Strong expertise in Python, SQL, PyTorch, Scikit-learn, MLflow, and AWS, with hands-on experience in predictive modeling, deep learning, NLP, RAG, scalable data pipelines, and MLOps. Experienced in developing production-ready ML applications, REST APIs, automated workflows, model deployment, monitoring, and cloud-based AI solutions.

## TECHNICAL SKILLS

**Programming & APIs:** Python, SQL, Bash/Shell Scripting, REST APIs, FastAPI, Git, Linux

**Machine Learning:** Scikit-learn, XGBoost, LightGBM, Classification, Regression, Clustering, Anomaly Detection, Predictive Modeling, Forecasting, Feature Engineering, Model Evaluation, Hyperparameter Tuning

**Deep Learning & NLP:** PyTorch, TensorFlow, Neural Networks, Transformers, Hugging Face, BERT, Natural Language Processing, Text Classification

**Generative AI:** LLMs, Prompt Engineering, Embeddings, RAG, Semantic Search, Vector Search, Document Q&A, LangChain

**MLOps:** MLflow, Apache Airflow, Model Deployment, Model Serving, Model Registry, Model Versioning, Model Monitoring, Experiment Tracking, CI/CD, Docker, Kubernetes

**Data Engineering:** ETL/ELT, Data Ingestion, Data Transformation, Data Validation, Data Quality, Apache Spark, PySpark, Apache Kafka, Hadoop, Databricks, Delta Lake

**Cloud & DevOps:** AWS, S3, SageMaker, Glue, Redshift, Docker, Kubernetes, Jenkins, GitHub Actions

**Databases:** PostgreSQL, MySQL, SQL Server, Oracle, Snowflake, MongoDB

**Data Analysis:** Pandas, NumPy, Matplotlib, Exploratory Data Analysis, Statistical Analysis

**Development Methodologies:** Agile, Scrum, SDLC, Code Reviews

## PROFESSIONAL EXPERIENCE

**AI/ML ENGINEER | LEAR CORPORATION** | Mar 2025 – Present | Southfield, Michigan

- Develop Python and SQL-based machine learning solutions for predictive analytics, classification, anomaly detection, and decision-support use cases.
- Build end-to-end ML pipelines covering data ingestion, validation, feature engineering, model training, evaluation, inference, and scheduled processing using Python, PySpark, Apache Airflow, and AWS.
- Perform exploratory data analysis and statistical analysis with Pandas and NumPy to identify predictive features, anomalies, correlations, and data-quality issues.
- Develop and evaluate classification and regression models using Scikit-learn and XGBoost, applying cross-validation and hyperparameter tuning to improve model performance.
- Develop and evaluate neural-network and deep-learning models using PyTorch and TensorFlow for predictive modeling and classification use cases.
- Implement feature engineering and selection workflows for missing-value treatment, categorical encoding, normalization, aggregation, and derived predictive features.
- Develop NLP and Generative AI applications using LLMs, Hugging Face, embeddings, RAG, semantic search, and prompt engineering for document-based question-answering use cases.
- Build document-processing workflows for parsing, metadata extraction, embedding generation, vector indexing, retrieval, and context construction using LangChain and vector search technologies.
- Develop FastAPI-based REST services to expose ML models and AI capabilities to downstream applications, including request validation, logging, exception handling, and testing.
- Use MLflow for experiment tracking, model versioning, artifact management, model registry, and ML lifecycle management.
- Automate ML workflows using Apache Airflow and CI/CD pipelines with GitHub Actions and Jenkins, containerize ML applications using Docker and support scalable deployment with Kubernetes.
- Use AWS S3, SageMaker, Glue, Redshift, and CloudWatch for cloud-based data processing, model development, deployment, monitoring, and production support; participate in Agile/Scrum, SDLC, code reviews, testing, and sprint activities.

**DATA ENGINEER | CAPGEMINI** | Jul 2021 – Dec 2022 | Bengaluru, India

- Developed Python and SQL-based data-processing solutions supporting analytics and machine learning workloads.
- Built ETL/ELT pipelines integrating relational databases, APIs, flat files, cloud storage, and enterprise data sources.
- Used Apache Kafka for data ingestion and streaming workflows supporting downstream data processing and analytics pipelines.

- Developed scalable data transformations using Python, SQL, Pandas, PySpark, and Apache Spark for structured and semi-structured datasets.
- Prepared machine learning datasets through data cleansing, deduplication, validation, normalization, transformation, and feature engineering.
- Performed exploratory data analysis using Pandas, NumPy, Matplotlib, and statistical techniques to identify patterns, trends, anomalies, and data-quality issues.
- Developed classification, regression, clustering, and anomaly-detection models using Scikit-learn for business and analytical use cases.
- Evaluated models using accuracy, precision, recall, F1-score, ROC-AUC, MAE, MSE, and RMSE, applied feature selection, cross-validation, and hyperparameter tuning.
- Built reusable Python modules and Apache Airflow workflows for data ingestion, preprocessing, validation, ML dataset preparation, and workflow automation.
- Processed large-scale datasets using Hadoop, PySpark, and Spark SQL while optimizing distributed transformations, joins, and partitioning and implemented data-quality checks for schema validation, missing values, duplicates, invalid records, data types, and pipeline completeness.

## EDUCATION

Master's Degree in Data Analytics – Indiana Wesleyan University | Aug 2023 – May 2025

Bachelor's Degree in Information Technology | July 2017 – June 2021

## CERTIFICATIONS

- AWS Certified Machine Learning Engineer – Associate
- Microsoft Certified: Azure AI Engineer Associate
- Databricks Certified Machine Learning Associate